



UITs

UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES

MACHINE LEARNING

Course Code: CSE-431

SUBMITTED TO

Mrinmoy Biswas Akash

Lecturer of University of Information Technology and Sciences (UITs)

SUBMITTED BY

Md. Shoyaif Rahman

ID: 0432310005101050

Batch: 53

Semester: Spring (2026)

Section: 7A

Date of Submission: 27/06/2026

Task 1: Performance Metric Analysis

Understanding Classification Metrics in an Industrial Context

In binary classification problems such as machine failure prediction, a model assigns each observation to one of two classes: Failure (positive class) or Normal (negative class). The fundamental building block for all performance metrics is the Confusion Matrix, which tabulates four outcomes: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Each metric captures a different facet of this performance landscape.

Accuracy:

Accuracy is the proportion of all predictions (both positive and negative) that are correct. It is expressed as:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

In the context of CNC maintenance, an accuracy of 90% (SVM) indicates that 90 out of every 100 machine-state predictions are correct. While seemingly impressive, accuracy is a misleading metric when class imbalance exists. Machine failures are rare events; if only 5% of all observations correspond to actual failures, a naïve classifier that always predicts 'Normal' would achieve 95% accuracy while detecting zero failures — a catastrophic outcome in an industrial setting.

Precision:

Precision (also called Positive Predictive Value) measures the proportion of predicted failures that are genuine failures:

$$\text{Precision} = TP / (TP + FP)$$

A precision of 0.91 (SVM) means that 91% of the maintenance alerts issued by the model correspond to real failures, with only 9% being false alarms. High precision is important for Operations Managers who wish to avoid unnecessary maintenance interruptions. However, pursuing precision at the expense of recall can result in genuine failures being missed entirely.

Recall (Sensitivity):

Recall, also known as sensitivity or True Positive Rate, measures the proportion of actual failures that the model successfully detects:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

A recall of 0.95 (DNN) indicates that the model successfully detects 95% of all actual machine failures. In predictive maintenance, recall is arguably the most safety-critical metric, as each missed failure (false negative) can result in \$100,000 in production losses, worker safety risks, and delivery delays. A recall of 0.82 (SVM) means that 18% of actual failures go undetected — an unacceptably high miss rate in a high-stakes industrial environment.

F1-Score:

The F1-Score is the harmonic mean of Precision and Recall, providing a single composite measure that penalises extreme imbalances between the two:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The DNN achieves the highest F1-Score of 0.91, indicating the best overall balance between detecting failures and avoiding false alarms. XGBoost follows closely at 0.89. The F1-Score is particularly valuable when stakeholders must jointly optimise for both safety (recall) and operational efficiency (precision).

Class Imbalance and Its Impact

Machine failure datasets are inherently imbalanced: failures represent a small fraction of total operating hours. In an environment where machines operate continuously, a failure event may constitute fewer than 2-5% of all recorded states. This imbalance causes accuracy to become misleading. Techniques such as Synthetic Minority Over-sampling (SMOTE), class weighting, and threshold tuning are essential preprocessing steps to ensure models do not develop a bias toward predicting the majority (Normal) class. Without addressing class imbalance, even a high-accuracy model may exhibit poor recall and fail to provide practical value in an industrial setting (He and Garcia, 2009).

Summary Comparison Table: Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
SVM	90%	0.91	0.82	0.86
XGBoost	92%	0.89	0.90	0.89
DNN	94%	0.87	0.95	0.91

TASK 2: COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS

Support Vector Machine (SVM)

A. Working Principle:

SVM is a kernel-based supervised learning algorithm that seeks to identify the optimal hyperplane separating the two classes (Failure vs. Normal) in a high-dimensional feature space. By applying kernel functions (Radial Basis Function, polynomial, or linear), SVM can model non-linear decision boundaries without explicitly computing the transformed feature space. The model is parameterised by support vectors — the training samples closest to the decision boundary — which define the maximum-margin classifier (Cortes and Vapnik, 1995).

B. Strengths

- High precision (0.91): minimises costly false maintenance alarms.
- Low inference time: ideal for near real-time edge deployment on IIoT gateways.
- Strong generalisation in small-to-medium datasets.
- Kernel trick enables non-linear classification without explicit feature engineering.
- Highest explainability among the three models, supporting regulatory compliance.

C. Limitations

- Recall of 0.82 is the lowest of the three models, meaning 18% of real failures go undetected.
- Training time scales poorly with very large datasets ($O(n^2)$ to $O(n^3)$ complexity).

- Sensitive to feature scaling and hyperparameter tuning.
- Not natively well-suited to temporal sequence modelling without handcrafted features.

D–G. Industrial Suitability and Risk Assessment

SVM is well-suited for environments where false alarms must be minimised and predictions must be generated rapidly at the edge. However, its recall deficit represents a significant risk: for every 100 actual failures, SVM misses approximately 18, each carrying a \$100,000 cost. This yields an expected undetected failure cost of \$1,800,000 per 100 failure events — a considerable financial and safety liability. SVM is most appropriate where dataset size is moderate and regulatory explainability is non-negotiable.

XGBoost

A. Working Principle

XGBoost (Extreme Gradient Boosting) is an ensemble learning method based on gradient boosting of decision trees. It constructs a sequence of shallow trees, each trained to correct the residual errors of its predecessors, with regularisation terms added to prevent overfitting. The final prediction is the weighted sum of all individual tree outputs (Chen and Guestrin, 2016). XGBoost has become a dominant algorithm in tabular data competitions due to its efficiency, robustness, and built-in handling of missing values.

B. Strengths

- Balanced precision (0.89) and recall (0.90) with the best overall F1-Score candidate.
- Natively handles missing sensor data and noisy features.
- Feature importance scores enable partial model interpretability (SHAP values).
- Moderate inference time is compatible with near real-time requirements.
- Regularisation (L1/L2) reduces overfitting on imbalanced datasets.
- Supports parallel computation for faster training.

C. Limitations

- Lower explainability than SVM: predictions are ensemble outputs rather than single function values.
- Hyperparameter tuning (learning rate, depth, number of estimators) is complex.
- Memory consumption increases with dataset size and tree depth.

D–G. Industrial Suitability and Risk Assessment

XGBoost represents a robust industrial choice. Its recall of 0.90 means that for every 100 failures, 90 are detected — 8 more than SVM. The cost saving is $8 \times \$100,000 = \$800,000$ per 100 events

relative to SVM. The slightly higher false alarm rate (precision 0.89 vs SVM 0.91) results in marginal additional maintenance costs that are easily offset. Medium deployment cost and partial explainability via SHAP analysis make XGBoost a technically and commercially viable option for most smart manufacturing contexts.

DEEP NEURAL NETWORK (DNN)

A. Working Principle

A Deep Neural Network is a multi-layer perceptron with numerous hidden layers, enabling hierarchical feature learning from raw input data. Each layer applies a linear transformation followed by a non-linear activation function (ReLU, sigmoid). Backpropagation with stochastic gradient descent (or adaptive optimisers such as Adam) trains the weights to minimise a cross-entropy loss function. DNNs can capture highly complex, non-linear patterns and temporal dependencies when augmented with recurrent or convolutional layers (LeCun et al., 2015).

B. Strengths

- Highest accuracy (94%) and recall (0.95): detects 95% of all actual failures.
- Capable of automated feature extraction, reducing reliance on domain expertise.
- Scalable: performance generally improves with more data.
- Architectures (LSTM, CNN-LSTM) can model temporal sensor patterns effectively.

C. Limitations

- Lowest explainability: often considered a 'black box', creating regulatory compliance challenges.
- Highest inference time, which may violate near real-time deployment constraints.
- Highest deployment cost due to GPU/TPU hardware requirements.
- Requires large training datasets to avoid overfitting.
- Computationally intensive training and tuning processes.

D–G. Industrial Suitability and Risk Assessment

DNN's high recall (0.95) is its most valuable attribute in this application. It would detect 95 out of 100 failures, missing only 5 at a cost of \$500,000 per 100 events — the lowest undetected failure cost of all three models. However, its 'black box' nature directly conflicts with safety regulations requiring explainable and auditable decisions. Additionally, its high inference latency may not meet the near real-time processing requirement, and its deployment cost is prohibitive for large-scale rollout across hundreds of machines.

Comprehensive Model Comparison Table

Criterion	SVM	XGBoost	DNN
Accuracy	90%	92%	94% ★
Precision	0.91 ★	0.89	0.87
Recall	0.82	0.90	0.95 ★
F1-Score	0.86	0.89	0.91 ★
Explainability	High ★	Medium	Low
Inference Time	Low ★	Medium	High
Deployment Cost	Low ★	Medium	High
Scalability	Low	High	High ★
Real-Time Suitability	High ★	Medium	Low
Regulatory Compliance	High ★	Partial	Low

TASK 3: ENGINEERING DECISION-MAKING AND TRADE-OFF ANALYSIS

Role as Lead AI Engineer

As Lead AI Engineer, the mandate is to select and justify a single model for deployment across the company's CNC machine fleet. This decision must balance predictive performance, safety, regulatory compliance, real-time processing requirements, and total cost of ownership. The analysis below presents a rigorous, evidence-based trade-off assessment structured around the specific engineering constraints of this deployment context.

Trade-Off 1: Accuracy vs. Explainability

The DNN achieves the highest accuracy (94%) but offers the lowest explainability. Safety regulations governing maintenance decisions in industrial environments typically require operators to understand and justify why an alert was triggered. A DNN's black-box nature prevents engineers and safety officers from auditing the decision logic, creating significant regulatory and liability exposure. XGBoost, with medium explainability supported by SHAP

(SHapley Additive exPlanations) values, enables feature-level attribution without sacrificing substantial predictive performance. This trade-off favours XGBoost as the operationally compliant choice.

Trade-Off 2: Recall vs. False Alarm Rate

The cost asymmetry between false negatives (FN = \$100,000) and false positives (FP = \$5,000) has a ratio of 20:1. This strongly incentivises maximising recall. However, excessively high false alarm rates erode operator trust and result in 'alert fatigue', whereby engineers begin ignoring alerts — ultimately defeating the purpose of the PdM system. XGBoost's precision of 0.89 and recall of 0.90 represent a commercially rational equilibrium: sufficient recall to avoid costly undetected failures, with a false positive rate that remains operationally manageable.

Quantitative estimate (per 100 actual failure events):

Model	Failures Detected	Failures Missed	Missed Cost	Net Risk Reduction vs Baseline
No Model (Baseline)	0	100	\$10,000,000	—
SVM (Recall 0.82)	82	18	\$1,800,000	\$8,200,000 saved
XGBoost (Recall 0.90)	90	10	\$1,000,000	\$9,000,000 saved
DNN (Recall 0.95)	95	5	\$500,000	\$9,500,000 saved

While DNN achieves an additional \$500,000 saving over XGBoost per 100 failures, these gains are partially offset by its higher false positive costs. If XGBoost generates 10 false alarms per 100 failure alerts (precision 0.89 $\approx$ 89 TP out of $\sim$ 100 alerts = $\sim$ 11 FP) and DNN generates 13 (precision 0.87 $\approx$ 87 TP out of $\sim$ 100 alerts = $\sim$ 13 FP), the additional FP cost for DNN is $2 \times \$5,000 = \$10,000$ — negligible compared to the broader cost drivers but indicative of DNN's slightly more alarm-prone behaviour.

Trade-Off 3: Predictive Performance vs. Deployment Cost

Deploying DNN across hundreds of CNC machines demands significant GPU-enabled infrastructure, specialised MLOps pipelines, and dedicated data science support. These costs — both capital and operational — may make enterprise-wide deployment economically unviable. XGBoost, by contrast, can be deployed on standard CPU-based servers or even at the edge on industrial PCs, at a fraction of the hardware cost. Given that the incremental performance

advantage of DNN is modest relative to XGBoost (2 percentage points in accuracy, 5 points in recall), the cost-effectiveness analysis strongly favours XGBoost.

Trade-Off 4: Safety vs. Operational Efficiency

Safety Officers prioritise zero missed failures; Operations Managers prioritise minimum downtime from false alarms. These objectives are inherently in tension. A threshold calibration approach can partially resolve this conflict: by adjusting the decision threshold of XGBoost below the default 0.5 (e.g., to 0.35), recall can be further increased at the cost of slightly more false alarms. This threshold tuning flexibility allows the engineering team to dynamically balance safety and operational efficiency without retraining the model — a practical advantage unavailable to fixed SVM configurations.

Trade-Off 5: Real-Time Performance vs. Model Complexity

The requirement for near real-time predictions is particularly critical when sensor data arrives in high-frequency bursts and decisions must propagate to maintenance crews rapidly. DNN's high inference time creates a risk of prediction latency during peak production, particularly in environments where multiple machines must be scored simultaneously. XGBoost's medium inference time is compatible with near real-time requirements when deployed on optimised inference engines. SVM, while fastest, sacrifices recall performance that renders it unsuitable as the sole safety mechanism.

Final Recommendation: XGBoost

Based on the multi-dimensional trade-off analysis presented above, XGBoost is recommended as the deployment model. The justification is as follows:

Performance: XGBoost achieves 92% accuracy, 0.90 recall, and 0.89 F1-Score — meaningfully superior to SVM and approaching DNN's performance while respecting operational constraints.

Cost Efficiency: XGBoost saves \$9,000,000 per 100 failure events versus the no-model baseline, compared to \$8,200,000 for SVM and \$9,500,000 for DNN. The marginal \$500,000 gap relative to DNN does not justify DNN's substantially higher deployment cost.

Regulatory Compliance: SHAP-based feature attribution provides sufficient explainability to satisfy regulatory and auditing requirements.

Real-Time Feasibility: Medium inference time is deployable on standard industrial hardware, meeting near real-time requirements without GPU dependency.

Stakeholder Alignment: XGBoost serves the diverse stakeholder matrix — Maintenance Engineers benefit from high recall; Operations Managers benefit from controlled false alarm rates; Financial Managers benefit from lower deployment cost; Safety Officers are served by strong failure detection; Regulatory Authorities are satisfied by SHAP explainability.

Deploying SVM alone would result in 18% of failures going undetected, causing avoidable losses of \$1,800,000 per 100 events. Deploying DNN without addressing explainability requirements risks regulatory non-compliance, audit failures, and loss of operator trust. XGBoost navigates these constraints most effectively, making it the industrially optimal choice.

TASK 4: REAL-WORLD DEPLOYMENT CHALLENGES

Challenge 1: Sensor Noise and Data Quality Issues

Industrial sensors operate in harsh physical environments characterised by electromagnetic interference, mechanical vibration, temperature extremes, and chemical exposure. These conditions introduce noise, outliers, and systematic drift into sensor readings. For example, a vibration sensor mounted near a cooling fan may capture spurious high-frequency signals unrelated to spindle bearing degradation. Similarly, a thermocouple with degraded insulation may report erratic temperature values that corrupt the input feature space.

The impact on model performance can be severe: noisy features reduce the signal-to-noise ratio of the training data, causing models to learn spurious correlations rather than genuine failure signatures. This manifests as degraded precision and recall on real deployment data, even if cross-validation performance on historical data appeared strong. Calibration drift can cause a well-trained model to become progressively less accurate over time as sensor behaviour departs from the training distribution.

Mitigation strategies include deployment of digital filters (Kalman, Butterworth bandpass) at the IIoT edge layer, implementation of automated anomaly detection to flag sensor outliers for exclusion, regular sensor calibration schedules tied to the PdM system's maintenance workflow, and ensemble feature imputation techniques for handling missing or corrupted readings. In industrial practice, companies such as Bosch and Siemens employ multi-sensor fusion architectures that cross-validate readings across redundant sensor channels before feeding data to prediction models (Vichare and Pecht, 2006).

Challenge 2: Class Imbalance in Machine Failure Datasets

Machine failure events are statistically rare. In a well-maintained fleet of CNC machines, failures may constitute as little as 0.5–5% of all recorded operational observations. This severe class imbalance causes standard classification algorithms to optimise for the majority class, producing models that achieve high accuracy by predicting 'Normal' for nearly all observations while exhibiting dangerously low recall for the minority Failure class.

The business consequences are significant: a model trained on imbalanced data without correction may appear to perform well in cross-validation (where accuracy is high) while being practically useless as a safety tool (where recall is critically low). This discrepancy between development-environment metrics and production outcomes represents one of the most common causes of ML deployment failure in industrial settings (He and Garcia, 2009).

Mitigation strategies include: (1) SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic failure examples by interpolating between existing minority-class samples; (2) Cost-sensitive learning by assigning higher misclassification penalties to false negatives during training (e.g., class weight of 20 for failure class consistent with the 20:1 cost ratio); (3) Threshold calibration to adjust the decision boundary post-training; (4) Anomaly detection as a complementary approach treating failures as outliers in the Normal operational distribution. XGBoost's `scale_pos_weight` hyperparameter provides a direct mechanism for cost-sensitive training in imbalanced settings.

Challenge 3: Model Interpretability and Explainability

Safety regulations, internal audit requirements, and operator trust all demand that maintenance decisions be explainable. When a model issues a failure alert, maintenance engineers, safety officers, and management must understand the causal rationale: which sensor reading triggered the alert, how confident the model is, and what physical deterioration mechanism the alert likely reflects. Without this transparency, operators are unable to verify alerts, risk making incorrect responses, and may lose confidence in the system entirely.

The explainability challenge is most acute for DNN models, which encode predictive logic across millions of interconnected weights that are individually meaningless. Even for XGBoost, the ensemble nature of the model makes it non-trivially interpretable without additional tooling. SVM provides the most naturally interpretable decision function but at the cost of lower recall.

Mitigation strategies include: (1) Application of SHAP (SHapley Additive exPlanations) values to attribute XGBoost predictions to specific sensor features, providing engineers with ranked lists of contributing variables for each alert; (2) LIME (Local Interpretable Model-agnostic Explanations) to generate locally faithful linear approximations of individual predictions; (3) Development of human-readable alert dashboards that translate model outputs and SHAP attributions into plain engineering language (e.g., 'Bearing temperature anomaly — spindle vibration exceeded 3-sigma threshold for 12 minutes'); (4) Audit logging of all model inputs,

outputs, and SHAP attributions for regulatory review (Ribeiro et al., 2016; Lundberg and Lee, 2017).

Summary Table of Deployment Challenges

Challenge	Impact on Model	Business Consequence	Mitigation Strategy
Sensor Noise & Data Quality	Degraded precision/recall; concept drift over time	Missed failures; false alarms; erosion of trust	Digital filtering; anomaly detection; sensor fusion; regular calibration
Class Imbalance	High accuracy but critically low recall; bias toward Normal class	Undetected failures; regulatory risk; model rejected in production	SMOTE; cost-sensitive learning; threshold calibration; anomaly detection
Model Interpretability	Black-box outputs; inability to audit decisions	Regulatory non-compliance; operator distrust; liability exposure	SHAP values; LIME; audit logging; human-readable alert dashboards

TASK 5: JUSTIFICATION OF COMPLEX ENGINEERING PROBLEM ATTRIBUTES

This section formally documents how this assignment satisfies the attributes of a Complex Engineering Problem (CEP) as defined by engineering accreditation frameworks, specifically the Washington Accord and relevant national engineering councils. The following attributes are addressed: P1 (Depth of Knowledge Required), P3 (Wide-Ranging or Conflicting Requirements), and P7 (Significant Consequences in a Range of Contexts).

P1 — Depth of Knowledge Required

Definition

P1 stipulates that a complex engineering problem requires knowledge and skills from the frontier of current engineering and computational science disciplines, extending beyond single disciplines and involving the application of advanced theory and methodology not typically addressed in foundational coursework.

Satisfaction of P1 in This Assignment

This assignment required the simultaneous application of advanced knowledge across multiple domains of engineering and computer science. Students must demonstrate mastery of statistical learning theory (SVM kernel methods, gradient boosting frameworks, deep learning

architectures), probabilistic reasoning (Bayesian interpretation of classifier metrics), industrial systems engineering (IIoT sensor data pipelines, CNC machine operation, maintenance management), regulatory and safety engineering frameworks, and engineering economics (cost-benefit analysis under uncertainty).

The integration of these knowledge domains to produce a coherent model selection recommendation — informed by quantitative cost analysis, regulatory constraints, and stakeholder requirements — represents frontier-level competency beyond routine undergraduate algorithmic knowledge. Furthermore, the requirement to apply explainable AI techniques (SHAP, LIME) to industrial decision systems reflects the current state of the art in applied machine learning research, which remains an active area of academic and industrial investigation (Molnar, 2019).

P3 — Wide-Ranging or Conflicting Requirements

This assignment presents an exceptionally rich landscape of conflicting requirements spanning technical, financial, operational, safety, and regulatory dimensions simultaneously. The five trade-offs systematically analysed in Task 3 demonstrate this characteristic:

Accuracy vs. Explainability: the highest-performing model (DNN, 94%) is the least explainable, directly conflicting with regulatory requirements.

Recall vs. False Alarm Rate: maximising failure detection (safety requirement) increases false alarms (operational disruption).

Predictive Performance vs. Deployment Cost: the model offering maximum failure prevention (DNN) requires the highest capital investment, conflicting with Financial Manager objectives.

Real-Time Performance vs. Model Complexity: deeper models require longer inference times, potentially violating real-time system requirements.

Safety vs. Operational Efficiency: zero-tolerance failure detection conflicts with minimal interruption to production.

P7 — Significant Consequences in a Range of Contexts

Definition

P7 requires that the problem carry significant consequences across multiple contexts — technical, economic, social, safety, environmental, or regulatory — such that engineering decisions carry substantial real-world impact extending beyond the immediate technical domain.

Satisfaction of P7 in This Assignment

The deployment of a predictive maintenance system in a smart manufacturing environment carries demonstrable significant consequences across each of the following contextual dimensions:

Technical: model selection directly determines the quality of the predictive signal guiding maintenance interventions. An incorrect choice results in systematic failure detection gaps or excessive false alarm rates, undermining the technical purpose of the system.

Economic: as quantified in Task 3, the cost differential between model choices amounts to hundreds of thousands of dollars per failure event. At scale across hundreds of machines operating continuously, the cumulative financial impact of model selection is material and commercially significant.

Safety: missed failure predictions create conditions for catastrophic machine failures, including spindle seizure, tool ejection, and associated worker injury risks. The safety consequences of low-recall models are not merely operational but potentially life-threatening.

Regulatory: safety regulations governing maintenance decision documentation require explainability and auditability. Non-compliance carries legal liability, potential plant shutdown orders, and reputational damage to the manufacturing organisation.

Social/Workforce: the deployment of AI-driven maintenance systems changes workforce roles, demanding new digital competencies from maintenance engineers and potentially displacing traditional maintenance scheduling jobs. Ethical deployment requires transparency and workforce engagement.

The breadth and magnitude of these consequences — spanning economic, safety, legal, and social domains — firmly establish this assignment as satisfying the P7 criterion for a Complex Engineering Problem of significant contextual impact.

CONCLUSION

This report has conducted a rigorous and multidimensional comparative analysis of Support Vector Machine, XGBoost, and Deep Neural Network models for predictive maintenance in a smart manufacturing CNC environment. The analysis addressed performance metric evaluation, model comparison, engineering trade-off reasoning, deployment challenge identification, and complex engineering problem justification.

Performance metric analysis revealed that accuracy alone is an insufficient criterion for model selection in the presence of class imbalance and asymmetric misclassification costs. Recall emerged as the most critical metric for failure prevention, while precision governs operational efficiency. The F1-Score provides a composite measure that must be interpreted alongside contextual engineering constraints.

Model comparison showed that DNN achieves the highest predictive performance (accuracy 94%, recall 0.95, F1 0.91) but is constrained by black-box explainability, high deployment cost, and elevated inference time. SVM offers the strongest explainability and real-time performance but carries an unacceptably low recall of 0.82, leaving 18% of failures undetected. XGBoost presents the most balanced profile across performance, explainability, cost, and operational feasibility.

The trade-off analysis, supported by quantitative cost modelling, demonstrated that XGBoost saves approximately \$9,000,000 per 100 failure events relative to no-model operation, approaching DNN's \$9,500,000 saving while avoiding DNN's regulatory, cost, and latency challenges. XGBoost is therefore recommended as the optimal deployment model.

Deployment challenges — sensor noise, class imbalance, and interpretability — each require dedicated engineering mitigation strategies to ensure sustainable model performance in production. The Complex Engineering Problem justification demonstrated that this problem satisfies P1 (depth of knowledge), P3 (conflicting requirements), and P7 (significant consequences) in full.

In conclusion, the successful deployment of machine learning-based predictive maintenance in smart manufacturing requires not only technical excellence in model development, but also engineering wisdom in balancing performance, safety, explainability, regulatory compliance, and economic feasibility. XGBoost, augmented with SHAP explainability tooling and robust data preprocessing pipelines, represents the industrially optimal solution for this context.